MILITARY SPECIFICATION

INDICATOR, GYROSCOPE, VERTICAL REFERENCE ID-1481(*)/A

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers design and all performance requirements of a self-contained attitude indicator for aircraft use.

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Item	Type Designation
Indicator, Attitude Self-contained	ID-1481/A

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification, to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, methods of
MIL-C-4150	Case, Carrying, Water-vaporproof
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for

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MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-T-18303	Test Procedures; Preproduction and Inspection for Aircraft Electronic Equipment, format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-L-25467	Lighting, Integral, Aircraft, General Specification for

STANDARDS

Federal

FED-STD-209	Clean Room and Work Station Requirements, Controlled Environment
FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of

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MS3112	Connector, Receptacle, Electric, Box Mounting
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of

2.1.1 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials fulfillment of major design objectives shall be the prime consideration. Parts and materials shall, where possible, conform to Specification MIL-E-5400.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

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3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 2.50 pounds.

3.3.2 Reliability -

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 500 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have total operating life of 5,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment shall have 1,500 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088 and Figure I. The external wiring shall be unshielded, except that a minimum of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

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3.3.4 Interchangeability of Re-ordered Equipment - For re-ordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

3.3.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.3.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400.

3.3.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient ($25^{o}C \pm 5^{o}C$)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 $\pm$ 1.0 V 3 Phase AC and 5.0 $\pm$ 0.5 V AC/DC

3.3.9 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment, except as modified herein:

3.3.9.1 Vibration - The indicator shall operate satisfactorily while being subjected to vibration requirements of Curve II of Specification MIL-E-5400.

3.3.10 Warm-Up-Time - The time required for the equipment to

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warm up prior to operation shall be kept to a minimum and shall not exceed 3.5 minutes under standard conditions and 5.0 minutes at extreme service conditions.

3.3.11 Primary Input Power Requirements - The power supply to the indicator shall be 400-cycle three-phase 115 v line-to-line with phase sequence ABC.

3.3.11.1 Electrical Power Variations - The indicator shall function satisfactorily with variations in voltage from 103 to 123 and in frequency between 380 and 420 cps, in addition to the transient variations specified in MIL-STD-704.

3.3.11.2 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 seconds during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.3.12 Wiring - The internal electrical wiring of the indicator shall be insulated from its case. The external wiring shall be in accordance with Figure I.

3.3.13 Insulation Resistance - The insulation resistance of the indicator must be 50 megohms or more when 500 volts DC is applied between isolated electrical circuits and between electrical circuits and the chasis.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The indicator shall provide accurate visual presentation of aircraft displacement in the roll and pitch axes which are referenced to gravity, without requiring inputs from any external source other than power supply.

3.4.1.1 Accuracy - The indicator, when connected and operated at standard conditions, shall provide a pitch and roll accuracy of 0.25 degrees maximum.

3.4.1.2 Performance at Environmental Extremes - Performance

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deviations at environmental extremes shall be permitted as specified in the detail requirements.

3.5 Detail Requirements -

3.5.1 Design and Construction - The indicator shall comprise a 115 v three-phase 400 cycle operated vertical gyro mounted in a case with its outer gimbal axis fore-and-aft, and its inner gimbal axis athwartships. The vertical gyro shall be referenced to gravity with compensation for acceleration resulting from turns and fore and aft velocity changes. A moveable roll index shall be fixed to the outer gimbal; the index shall move over a scale fixed to the case. A moving drum, mechanically driven by the inner gimbal, shall register movement in the pitch and roll axes with respect to a miniature airplane mounted on an adjustable bracket and located at the center of the indicator glass cover. The attitude presentation shall be 360 degrees roll, 92 degrees climb and 78 degrees dive. The indicator shall be adjusted to indicate zero pitch when mounted with the case at an angle of 7 degrees from the vertical with the connector end of the indicator rotated downward. Rotation of a knob, located on the bezel, shall adjust the position of the miniature airplane to a maximum of five degrees in dive or climb. Pulling the knob shall cage the gyro in pitch and roll; rotating the knob clockwise while it is fully out shall lock the gyro in the caged position. A warning flag shall come into view when the gyro is caged, or if the AC power supply is interrupted. The indicator shall withstand the temperature extremes, the strains, jars and vibrations incident to shipping, storage installation and service.

3.5.1.1 Lighting - The attitude indicator shall be provided with integral lighting in accordance with MIL-L-25467.

3.5.1.2 Indicator Display - The display shall be in accordance with Figure 2, and shall consist basically of a miniature airplane, a bank angle dial, bank index, and a moving two-colored drum background with a distinct horizon line dividing the two colors. Unless otherwise specified, all colors shall be in accordance with FED-STD-595, with all visible portions of the indicator case being black Color No. 37038. The finish shall diffuse light sufficiently to avoid "HIGH-LIGHT" when illuminated with external or internal light sources.

3.5.1.2.1 Graduations, Lines - All markings of the indicator display shall be clean and sharp. Dimensions shall be as specified

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and on any given mark specified. All markings which for some given indication, become adjacent and which are specified to have the same basic dimension at the adjacent points shall not be noticeably different at that point.

3.5.1.2.2 Miniature Airplane - The miniature airplane shall be in accordance with Figure 6. The rear edge of the miniature airplane shall be within 1/16 inch of the drum surface.

3.5.1.2.3 Bank Angle Dial - The bank angle dial shall be in accordance with Figure 4. The scale markings of the dial shall be white Color No. 37875 of FED-STD-595.

3.5.1.2.4 Bank Index - The bank index shall be in accordance with Figure 5. It shall be affixed to rotate with the roll gimbal. The index shall not move with pitch deviations. The bank index shall be within 1/32 $\pm$ 1/64 inch of the small diameter edge of the scale surface of the bank angle dial. At zero roll indication the bank index shall be aligned with and radially within 1/32 $\pm$ 1/64 inch of the zero roll marking on the bank angle dial. The index shall be white Color No. 37875 of FED-STD-595.

3.5.1.3 Drum - The drum shall be mounted on the roll gimbal and shall rotate with the gimbal about the roll axis for roll indications, and rotate in relation to the roll gimbal about the pitch axis for pitch indications. The roll axis shall be perpendicular to the face of the indicator, the pitch axis shall be horizontal when the indicator is level and is indicating a zero roll attitude. The horizon line on the drum shall be at the zero pitch position. The zero pitch position shall be indicated when the case is inclined, with the connector end down, at an angle of 7 degrees from the horizontal. The drum shall be driven in pitch in such a way that for a dive maneuver the horizon line goes above the miniature airplane and for a climb maneuver the horizon line goes below the miniature airplane. The drum shall be in accordance with Figure 7. The surface of the drum shall move .026 (nominal) inch for each degree of pitch change.

3.5.1.3.1 Marking and Colors - The drum shall be marked as shown in Figure 7. Above the horizon line the drum shall be gray Color No. 36440 and all markings shall be black Color No. 37038. The horizon line shall be white Color No. 37875. Below the horizon line the drum shall be black Color No. 37038 and all markings shall be white Color No. 37875. The lengths of the major

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pitch lines shall be as follows:

0° (Horizon Line	Width of Drum)
10°	15/16 inch
20°	7/8 inch
30°	13/16 inch
40°	3/4 inch
50°	11/16 inch
60°	5/8 inch
70°	9/16 inch
80°	1/2 inch

The center of the dots shall be the extreme pitch position and from center to center shall represent 180 degrees of pitch indication. The pitch indication shall be linear over the full range and shall be spaced in accordance, Figure 7.

3.5.1.3.2 Numerals and Letters - All numberals and letters on the indicator shall be in accordance with MS33558.

3.5.1.4 Attitude Actuation - The drum shall be rotated about the pitch axis by means of gears from the vertical gyro inner gimbal. The drum and bank index shall be mounted directly on the outer gimbal of the vertical gyro. The freedom of the drum shall be unlimited about the roll axis and shall be 78 degrees dive and 92 degrees climb.

3.5.1.5 Pitch Trim - Means shall be incorporated to adjust the position of the miniature airplane by means of a pitch trim knob mounted on the lower right hand side of the face of the indicator and marked in accordance with Figure 2. The reference pointer on the knob shall align with the index scale on the case at the zero pitch trim position. The body color of the knob shall be black Color No. 37038, the reference arrow on the knob shall be white Color No. 37875, and the reference index scale on the case shall be white Color No. 37875. The knob shall have approximately

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one-eighth of a turn freedom in the clockwise direction and one-eighth of a turn freedom in the counter clockwise direction from the zero pitch trim position. The freedom in each direction shall be limited by a positive stop. When the knob is rotated clockwise to the stop, the miniature airplane shall move upward to indicate a 5 degree climb. When the knob is rotated counter clockwise to the stop the miniature airplane shall move downward to indicate a 5 degree dive. The knob shall have sufficient friction to remain fixed when set but it shall move smoothly without evidence of binding or erratic motion.

3.5.1.6 Caging - Means shall be incorporated to mechanically cage the gyroscope by pulling the pitch trim knob. With the knob fully extended and rotated to the extreme clockwise position, the gyroscope shall remain caged and the knob shall remain locked in the extended position. When the knob is pulled and power is applied, the power failure flag shall appear.

3.5.1.7 Power Failure Indication - A power failure flag shall be incorporated on the left of the face of the indicator. The design shall be such that the warning flag will appear when any or all phases of the power input is interrupted and when the gyro is caged. The warning flag shall be white Color No. 37875, and the lettering "OFF" shall be black Color No. 37038. The white warning flag shall be illuminated under night lighting conditions and the lighting shall not be entirely dependent upon one light source.

3.5.1.8 Cover Glass - The cover glass of the attitude indicator shall be flat and free from discolorations, scratches and striae which interfere with normal indicator reading.

3.5.1.9 Gimbal Freedom - Unlimited freedom shall be provided about the roll axis and approximately 92 degrees in climb and 78 degrees in dive. During power interruption, normal freedom is maintained in both axes. Pitch stops shall be incorporated which produce a roll precession when pitch freedom is executed. The pitch stop shall be so designed that no parts of the instrument will be damaged by a loop at the rate of 90 degrees per second and less than $\pm$ 4 degrees error will result from loops of 10 $\pm$ 1 degrees per second. Errors induced by roll maneuvers at a rate of 300 degrees per second shall not exceed 1 degree.

3.5.1.10 Anti-Spin Device - The indicator shall incorporate a device designed to prevent damage resulting from spinning of the

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outer gimbal in the absence of power.

3.5.1.11 Erection - An erection system which will maintain the alignment of the spin axis to the local vertical shall be provided. The erection system consists of a mechanical weight shift device driven directly from the gyro spin motor. The erection rate is reduced automatically when a false vertical in excess of 7 degrees occurs from turn or fore and aft acceleration.

3.5.1.11.1 Initial Erection - The caging knob shall erect the gyro to within 1 degree of the zero index from any position, after power is applied.

3.5.1.11.2 Final Erection - The gyro shall erect to within 0.5 degrees of vertical within 5 minutes after release of the caging knob. Power shall be applied for a minimum of 50 seconds before the final release of the caging knob.

3.5.1.11.3 Erection Rate - The gyro shall be capable of erecting in pitch and roll at a rate of 1.9 to 3.1 degrees per minute when the spin axis displacement does not exceed 7 degrees in pitch and roll. The erection rate in pitch and roll shall not exceed 1.5 degrees per minute when the spin axis displacement exceeds 10 degrees in pitch or roll.

3.5.1.12 Coast Time - The indicator shall display useful pitch and roll information for a minimum of 9 minutes after power is removed.

3.5.1.13 Case - The outline dimensions of the remote attitude indicator case shall conform to Figure 3.

3.5.1.14 Electrical Connector - The electrical connector used shall be in accordance with MS3112E10-6P. The electrical connections to the connector shall be in accordance with Table I of this specification.

TABLE I

PIN	CIRCUIT
A	115 v, 400 cps 3 phase AC power

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(Table I continued)

PIN	CIRCUIT
B	115v, 400 cps Phase rotation ABC
C	115v, 400 cps
D	Internal lighting 0 to 5 volts maximum
E	Internal lighting AC or DC
F	Spare

3.6 Controlled Environment - The requirements of FED STD 209 for clean room and work station environment shall be complied with.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made

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on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperatures shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract, and as a minimum shall consist of all the tests specified herein under Test Methods and the Life Test of 4.5.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government,

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it shall be the obligation of the contractor to make necessary corections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports

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showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Reliability Assurance Tests
(4) Special Tests

4.4.1 Individual Tests - Each indicator submitted for acceptance shall be subjected to and pass the following individual tests:

a.	Examination of product	(4.8.1)
b.	Initial erection	(4.8.2)
c.	Final erection	(4.8.3)
d.	Caging knob locking	(4.8.4)
e.	Power warning flag	(4.8.6)
f.	Gimbal freedom	(4.8.7)
g.	Anti-spin brake	(4.8.8)
h.	Pitch trim	(4.8.5)
i.	Erection rate	(4.8.9)
j.	Erection cut-off angle	(4.8.10)
k.	Visibility of Display	(4.8.29)
l.	Power Consumption	(4.8.11)
m.	Rotor coast time	(4.8.12)

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with the following: (Sampling tests shall not be

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conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action).

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

*NOTE: If by contract action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Attitude Indicator Sampling Tests - Each system selected for sampling tests shall be subjected to the following tests:

(a)	Power and Frequency Variation	(4.8.14)
(b)	Low Temperature Operation	(4.8.15)
(c)	High Temperature Operation	(4.8.16)
(d)	Magnetic Effect	(4.8.17)
(e)	Loop Error	(4.8.30)
(f)	Vibration Error	(4.8.18)

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Sampling Phase shall be conducted.

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4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level F shall be used, except that temperature extremes shall be +55°C and -44°C, and vibration shall be at 0.5G. The Accept-Reject Criteria for Test Plan IV shall be used.

4.4.3.2 Reliability Sampling Phase Tests - Samples of the equipment shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled "Sampling Phase of Production Reliability Tests". For the Sampling Phase, Test Level F shall be used, except that the temperature extremes shall be +55°C and -40°C, and vibration shall be 0.5G.

4.4.3.2.1 Lot Size for Sampling Phase - Each equipment produced shall be tested for 100 hours. In addition to the 100 hours a burn-in period may be used at the option of the contractor. If used, the burn-in must be the same for each equipment, in time and severity. To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan VI of MIL-STD-781. (Extend the line as necessary to accomodate the data). These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attn: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan VI curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.3.3 Test Details - The test details, such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristic to be measured, special failure criteria, preventive maintenance to be allowed during the test etc., shall be part of the test procedures to be submitted and approved by the

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procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment, and to assure adequate quality control. The equipment to be subjected to special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early indicator after a material or engineering change.
(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. Where such tests are not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance, or special tests, the following action shall be taken:

(1) Determine the cause of failure.
(2) Determine if the failure is an isolated case or design defect.
(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.
(4) Where practical, include the test in an individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

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4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing this test. The test shall be for 1000 hours duration and shall be conducted on equipments that have passed the individual tests. This test shall also be conducted as part of the preproduction test. The test shall be conducted under the conditions specified in 4.5.1 and Table IV. This test shall be conducted according to the following schedule. When reliability tests are conducted the life test shall be omitted if, during reliability tests, a quantity of equipments equal to more than that listed below receive at least 1000 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
Next 500	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency) three phase

TABLE IV

CONDITIONS	TEST CONDITION	TIME (APPROX. HOURS)
1	3 (±1-1/2) Total Angle, Pitch, Roll, and Yaw Motion	300
2	15 (±7-1/2) Total Angle, Pitch, Roll and Yaw Motion	300

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3	Vibration - 0.002 to 0.005 inch double amplitude and a frequency of 1500 to 2000 vibrations per minute	300
4	Tests conducted in accordance with this specification as specified by the procuring activity	100

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.3.1 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer
(5) Nature of the failure
(6) The number of hours which the part operated prior to failure

4.5.3.1.1 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occured. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.3.2 Reconditioning of Life Test Samples - An equipment which has been subjected to the Life Test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.
(2) After reworking, the contractor shall resubmit the equipment for acceptance.

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4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or contract. Specification MIL-R-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary to determine compliance with the requirements of this specification or the contract. The test methods of 4.8 are a minimum requirement.

4.7 Test Conditions -

4.7.1 Atmospheric Conditions - Unless otherwise specified, all tests required by this specification shall be made in accordance with the requirements of 3.3.9.

4.7.2 Attitude - Except as otherwise specified below, all tests shall be performed under the following conditions:

(a) The indicator shall be mounted and mechanically secured to a fixture so that the longitudinal axis of the indicator is inclined at an angle of 7° to the horizontal, with the connector end down. Wherever the term "Indicator" is used in the following paragraphs, it shall be understood to refer to the indicator mounted on the fixture as described above.

(b) All visual tests of indicator performance shall be observed from a distance of 2 feet in front of the glass cover, along the center line of the indicator.

4.7.3 Input Power - The input power shall be 115 $\pm$ 2 percent VAC, line to line 3 phase. AC power frequency shall be 400 $\pm$ 5 cps.

4.7.4 Test Equipment and Connections - Special test equipment and connections thereto shall be as specified. Whenever practical, the test equipment used in performing the tests shall not affect

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the performance of the equipment under test.

4.7.5 Test Readings - Unless otherwise specified, before a test reading is taken, the indicator shall, if necessary, be vibrated using a vibrator set at 30 cps, 60 cps, or 120 cps with a maximum amplitude of 0.002 inch.

4.7.6 Test Warm-Up-Time - For all tests except initial and final erection tests, a ten minute warm-up period shall be allowed before commencing any test.

4.8 Test Method -

4.8.1 Examination of Product - The indicator shall be examined carefully to determine that the material and workmanship requirements have been met.

4.8.2 Initial Erection - The indicator shall be mounted on a surface level within 1 minute of arc. Power shall be applied for a minimum of 50 seconds. Between 50 and 60 seconds the caging knob shall be pulled until the horizon line is stabilized. After returning the knob quickly, but avoiding a snap release, the indicator shall indicate level within 1 degree.

4.8.3 Final Erection - After the initial erection period, the indicator shall be mounted on a pitch, roll and yaw mechanism leveled within 0.5 degrees, displacing the gyro 1.5 degrees in pitch, roll and yaw, at a frequency 5 to 7 oscillations per minute. After 2 minutes of operation on this mechanism, the indicator shall be mounted on a surface level within 1 minute of arc, and shall indicate vertical within 0.5 degrees. When the indicator is rotated 180 degrees in azimuth the change in pitch and roll shall not indicate a change in verticality of the gyro spin axis of more than 1 degree.

4.8.4 Caging Knob Locking - The caging knob shall be pulled and held in the fully extended position until the horizon line is stabilized. The knob shall then be rotated to the extreme clockwise position; the indicator shall remain caged and the knob shall remain locked in the extended position when it is released.

4.8.5 Pitch Trim -

4.8.5.1 Pitch Trim Zero - With the indicator mounted on a

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level surface and power applied the unit shall be caged and quickly released. With the reference pointer on the knob aligned with the center index scale mark on the bezel the miniature airplane shall be aligned within ± 1/2 of the horizon line width.

4.8.5.2 Pitch Trim Adjustment - When the knob is rotated to the stop in the clockwise direction, the miniature airplane shall move upward to indicate a minimum of 5 degrees climb. When the knob is rotated to the stop in the counter clockwise direction, the miniature airplane shall move downward to indicate a minimum of 5 degrees dive.

4.8.6 Power Warning Flag -

4.8.6.1 Power Interruption - After a minimum of five minutes of operation the power shall be removed from pins A, B, and C in sequence. The power warning flag shall appear when power is removed from each pin and shall disappear when power is restored.

4.8.6.2 Low Voltage - When the 3 Ø 115 volt 400 cps power is decreased to 103 volts the flag shall not appear.

4.8.6.3 Caging Indication - With normal power applied, when the caging knob is pulled a maximum of .020 inches the flag shall appear. When the knob is released slowly a minimum of 0.008 inches travel shall exist after the flag has disappeared.

4.8.7 Gimbal Freedom - With the indicator operating normally it shall be displaced 360 degrees in roll, 92 degrees in climb and 78 degrees in dive. The drum and roll index shall follow smoothly without noticeable sticking, overshooting, hunting or binding.

4.8.8 Anti-Spin Brake - With normal power applied and the caging locked the indicator shall be positioned to a dive of 83 degrees. With the indicator at this position the caging knob shall be released and power removed. The indicator shall then be returned to the normal position. The rotational speed of the roll gimbal shall be measured after 30 seconds and shall not exceed 350 RPM.

4.8.9 Erection Rates - These tests should preferably be conducted with the indicator mounted on a pitch, roll and yaw mechanism with a movement of ± 1.5 degrees.

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4.8.9.1 Low Angle Erection -

4.8.9.1.1 Low Angle Erection - Roll Axis - With the indicator oriented towards the geographical West the gyro spin axis shall be inclined 7 degrees towards the geographical North. The time required for the gyro to erect to within 1 degree of the vertical shall be between 1 minute 56 seconds and 3 minutes 11 seconds. The pitch indicator shall not change more than 3.5 degrees during this test.

4.8.9.1.2 Low Angle Erection - Pitch Axis - With the indicator oriented towards the geographical North the gyro spin axis shall be inclined 7 degrees towards the North. The time required for the gyro to erect to within 1 degree of the vertical shall be between 1 minute 56 seconds and 3 minutes 11 seconds. The roll indication shall not change more than 3.5 degrees during this test.

4.8.9.2 High Angle Erection -

4.8.9.2.1 High Angle Erection - Roll Axis - With the indicator oriented towards the geographical West the gyro spin axis shall be inclined 30 degrees towards the North. The roll and pitch indication shall not change more than 6 degrees in a 4 minute period. The test shall be repeated with the gyro spin axis inclined 60 degrees towards the North. The same results shall be obtained.

4.8.9.2.2 High Angle Erection - Pitch Axis - With the indicator oriented towards the geographical North the gyro spin axis shall be inclined 30 degrees towards the North. The roll and pitch indication shall not change more than 6 degrees in a 4 minute period.

4.8.10 Erection Cut-Off Angle - With the indicator oriented towards the geographical North the gyro spin axis shall be inclined 12 degrees toward the North. The time required for the gyro to erect each degree down to 4 degrees shall be recorded. The recorded times shall show a marked increase in erection rate at an angle between 7 and 10 degrees.

4.8.11 Power Consumption - With normal voltage applied the current in each phase shall not exceed the following:

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0.200 amps starting
0.042 amps after 10 minutes of operation

4.8.12 Rotor Coast Time - With the indicator operating a minimum of 10 minutes the power shall be removed. The gyro shall not tumble for a minimum of 9 minutes.

4.8.13 Insulation Resistance - With a potential of 500 volts direct current applied for 5 seconds between isolated circuits and between circuits and the case, the resistance shall be greater than 50 megohms.

4.8.14 Power and Frequency Variation - The indicator shall be properly connected and the input voltage and frequency shall be varied within the following limits:

A. C. Volts	Frequency (cps)
103 to 127	420
103 to 127	380
103	380 to 420
127	380 to 420

The indicator shall then be subjected to and shall meet the requirements of 4.8.2, 4.8.3, 4.8.6, 4.8.9, 4.8.10, and 4.8.12.

4.8.15 Low Temperature Operation - The indicator shall be properly connected except that no power shall be applied. It shall then be tested in accordance with Step 2, Temperature and Altitude test, of MIL-T-5422, except as modified below, for 4 hours. At the end of this four-hour period and while at this reduced temperature, power shall be applied for 10 minutes. At the end of this 10-minute period, but still at the reduced temperature, the indicator shall be subjected to and shall meet the requirements of 4.8.2 initial erection, 4.8.3 final erection, and 4.8.9 erection rate (except erection time as measured shall be between 1½ minutes and 4 minutes 50 seconds). The indicator shall then be returned to standard conditions and allowed to stabilize for a period of 4 hours after which it shall be operated and shall meet the requirements of the individual tests. There shall be no damage which would

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adversely affect subsequent operation.

4.8.16 High Temperature Operation - The indicator shall be properly connected and tested in accordance with Step 6, Temperature and Altitude Tests, of MIL-T-5422. At the end of the 4-hour period, and while still at the elevated temperature, the indicator shall be subjected to and shall meet the requirements of the following: 4.8.2 initial erection, 4.8.3 final erection, and 4.8.9 erection rate (except erection time shall be between 1½ minutes and 4½ minutes). The indicator shall then be returned to standard conditions and allowed to stabilize for a period of 4 hours, after which it shall be subjected to and shall meet the requirements of the individual tests.

4.8.17 Magnetic Effect - The indicator shall be properly connected and power applied. The indicator shall be revolved about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from the bar magnet. The compass shall have its compensating magnets removed and shall be set up in a uniform magnetic field whose horizontal intensity is between 0.17 and 0.19 oersted. The indicator shall be revolved in a horizontal plane which is perpendicular to the axis of the bar magnet. The indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270, 315 degrees. At each of these positions the indicator shall be rotated 360 degrees about its horizontal axis. The deflection of the compass at any of the specified positions shall not exceed 5 degrees. This test shall be repeated with no power applied. The same tolerance shall apply.

4.8.18 Vibration Error - The indicator, properly connected, shall be mounted and vibrated for one hour as specified in Specification MIL-E-5272, Procedure IV. The diameter of circular motion shall be .003 to .005 inch and the slow sweep frequency rate shall be 5 $\pm$ 1 cps per minute. During this vibration period the indicator shall meet the requirements of 4.8.18.1 and 4.8.18.2.

4.8.18.1 Roll and Pitch Oscillation - The maximum total amplitude of roll and pitch oscillation throughout the frequency range shall be noted and shall not exceed 1 degree in roll and pitch.

4.8.18.2 Roll and Pitch Variation - The maximum variation in roll and pitch during vibration, from the original indication

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before vibration, throughout the frequency range, shall be noted and shall not exceed 2 degrees in roll and pitch.

4.8.19 Vibration Failure - The indicator shall be tested in accordance with Part II, Vibration Tests of MIL-T-5422, with power applied. At the conclusion of this test the equipment shall be subjected to and shall meet the requirements of the Individual Tests.

4.8.20 Radio Noise Interference - The indicator properly connected shall be tested for conducted and radiated radio noise interference in accordance with MIL-I-6181, except during switching of the power failure flag by pulling the caging knob.

4.8.21 High Temperature Exposure - The indicator shall be tested in accordance with Step 6, Temperature and Altitude Tests, of MIL-T-5422, except as modified for a period of 24 hours. The system shall be properly connected but no power shall be applied during the test. During the last 15 minutes of the 24-hour period power shall be applied. At the end of the 24-hour period and while still at the elevated temperature the equipment shall be subjected to and shall meet the requirements of 4.8.2 initial erection, 4.8.3 final erection, and 4.8.9 erection rate (except erection time shall be between 1½ minutes and 4½ minutes). The indicator shall then be returned to standard conditions and shall be allowed to stabilize for a period of 4 hours, after which it shall meet the requirements of the Individual Tests.

4.8.22 High Altitude - Low Temperature - The indicator properly connected but with no power applied, shall be subjected to Step 3, Temperature and Altitude Tests, of MIL-T-5422 except as modified for a period of 72 hours. At the end of the exposure period, power shall be applied for a minimum of 30 seconds. The chamber shall be opened and the caging knob pulled until the horizon line is stabilized. The chamber shall then be closed and returned to altitude pressure. The indicator shall be observed for 30 minutes and shall indicate the vertical within ± 2 degrees. After the equipment has returned to room temperature and pressure it shall be subjected to and shall meet the requirements of the individual tests.

4.8.23 Temperature Shock - The indicator shall be subjected to the Temperature Shock Test, Procedure I of MIL-E-5272. At the conclusion of this test the equipment shall be subjected to and

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shall meet the requirements of the Individual Tests.

4.8.24 Humidity - The indicator, with external connections to simulate installed conditions shall be mounted in its normal operating position in a chamber and maintained at a temperature of 70 ± 2 degrees C, (158 ± 3.6 degrees F.) and a relative humidity of 95 ± 5 percent for a period of 6 hours. Distilled water shall be used to obtain the required humidity. At the end of the 6-hour period, the heat shall be shut off and the indicator shall be allowed to cool for a period of 18 hours in this atmosphere, in which the humidity rises to 100 percent as the temperature decreases. This cycle shall be repeated a sufficient number of times to extend the total time of test to 120 hours (5 cycles). At the end of the 120-hour (5 cycle) period and within one hour after the indicator has been removed from the chamber, it shall be subjected to and meet the requirements of the Individual Tests. Examination for corrosion and other deterioration shall be made between 24 and 48 hours after completion of the humidity exposure. There shall be no evidence of corrosion or other deterioration which will affect subsequent operation of the indicator.

4.8.25 Salt Spray - The indicator shall be tested in accordance with paragraph 4.6.1 (Salt Spray Tests, Procedure I) of Specification MIL-E-5272. There shall be no damage to the indicator and no sticking or binding shall occur in the caging knob as a result of this test. At the end of the exposure period, the indicator shall meet the requirements of all the Individual Tests.

4.8.26 Fungus Resistance - The indicator shall be tested in accordance with paragraph 4.8.1 (Fungus Resistance Tests, Procedure I) of Specification MIL-E-5272, except that the test period shall be 14 days. No sticking or binding of the caging knob shall occur as a result of the test; there shall be no deterioration that would adversely affect the subsequent operation of the indicator, and fungus growth shall not be supported.

4.8.27 Impact Shock - The indicator shall be tested in accordance with the Shock Test of MIL-T-5422. At the end of this test, the equipment shall be subjected to and shall meet the requirements of the Individual Tests.

4.8.28 Acceleration - The indicator shall be subjected to 20g acceleration for a period of 1.0 minute along each of its 3 axes in both directions while not operating. At the conclusion of the test the equipment shall be subjected to and shall meet the requirements of the Individual Tests.

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4.8.29 Visibility of Display - The pitch scale graduations from 20 to -20 degrees, the bank index, the numerals and other specified markings on the dail shall be visible from any point within the frustum of a cone whose side makes an angle of 30 degrees with a perpendicular to the spheroid and whose small diameter is the aperture of the instrument case.

4.8.29.1 Illumination of Display - The indicator shall be integrally lighted in accordance with requirements of Specification MIL-L-25467.

4.8.29.2 Prime Standard - Using parts which are identical to the parts to be used in production indicators, an indicator shall be constructed which meets all display requirements of Specification MIL-L-25467. The indicator model and recorded lighting data shall be submitted to the procuring activity for approval; and if approved, the indicator model shall be certified as a prime standard.

4.8.29.3 Prime Standard Life - A minimum of 12 average ageas covering all representative areas of the face of the prime standard shall be measured for brightness and color prior to submission, and at least every 50 hours of operation thereafter. A change of +/-20% brightness or +/-5% color change in any area using the initial data as a reference voids the prime standard certification.

4.8.30 Loop Error - The indicator shall be mounted on a fixture permitting rotation about the pitch axis. The gyro shall be rotated one complete revolution (loop) at a rate of 10 $\pm$ 1 degree per second, and returned to the normal position. The error in either pitch or foll shall not exceed 4 degrees.

4.9 Reconditioning of Tested Equipment - Equipment which as been subjected to acceptance and longevity tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.10 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.11 Rejection and Retest - Equipment which has been rejected may be reworked to have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 Packaging - The indicator shall be packaged in accordance with MIL-STD-794, Level A. The method of preservation shall be in accordance with MIL-P-116, Method IC, utilizing a rectangular container. The container shall conform to MIL-C-4150.

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5.2 Packing - The indicator shall be packed in accordance with MIL-STD-794, Level A. The package and exterior shipping container may be one and the same container.

5.3 Rough Handling Tests - Rough handling tests shall apply to all levels of packaging and packing. The tests performed on a prototype example shall be Rough Handling, Dummy Load and Pre-production Tests of MIL-STD-794. No vibration tests are required. Two pilot shipping containers are required to be tested. The heights of drop shall be 30 inches for the free-fall flat drop and 36 inches for the free-fall cornerwise drop. All testing shall be done on all axes and in each direction in each axis and with actual indicator as packaged for shipment. After tests, the indicator shall not suffer damage or subsequently fail to provide performance as specified. All testing shall be done at the contractor's plant. The contractor shall notify the Naval Air Systems Command in sufficient time in advance of prototype testing to permit Government representation at the test site.

5.4 Marking - The interior and exterior containers shall be marked as specified in MIL-STD-129 as follows:

5.4.1 Precautionary Marking - The following precautionary markings warning of the fragility of the item shall appear on two opposite sides of each interior package and shipping container. Containers shall be marked with arrows on two opposite sides of each interior package and shipping container to indicate which surface must always be up:

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

UP



In addition, containers shall be marked "Reusable, OPEN WITH CARE' in letters at least 2 inches high, and on two opposite sides of each container.

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6. NOTES

6.1 Intended Use - The attitude indicator defined by this Specification is intended for use under the conditions laid down in Specification MIL-E-5400 for Class 2 equipment, as a flight reference instrument indicating aircraft displacement in the roll and pitch axes.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number and date of this Specification.
(2) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Precedence of Documents - When the requirements of the contract, this Specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.
(2) This Specification - This Specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.
(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and

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processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accomplished by complete supporting information.

6.5 The parentheses (*), when used in the type designation, will be deleted or replaced by either a number of letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.7. The complete type number shall be used in nameplates, shipping records and instruction books, as applicable.

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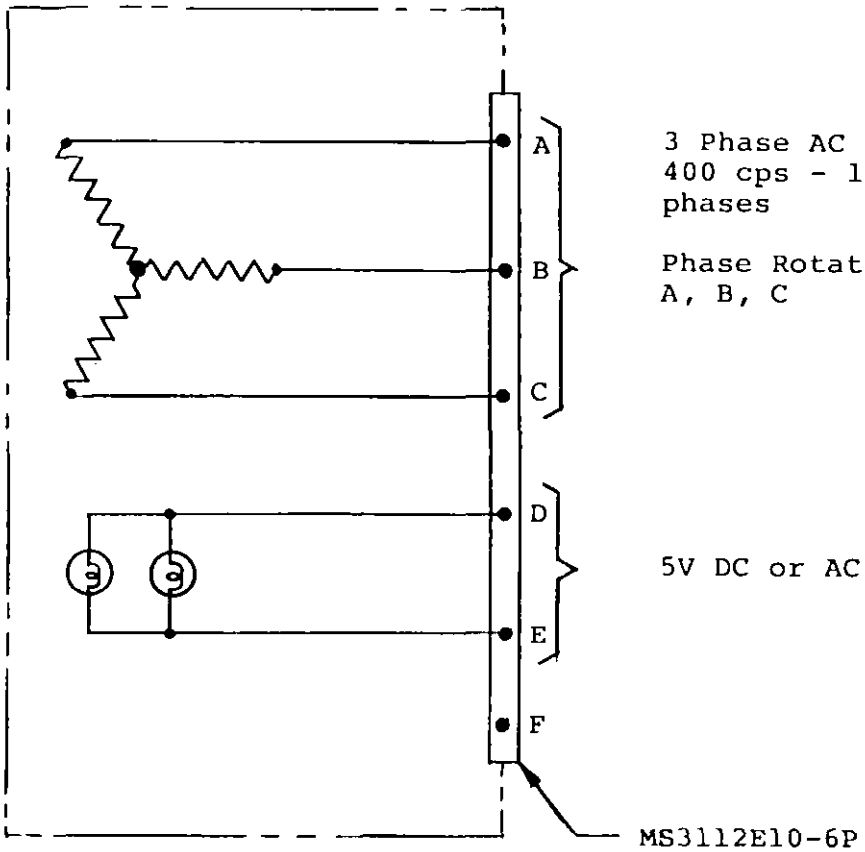


FIGURE 1 SCHEMATIC

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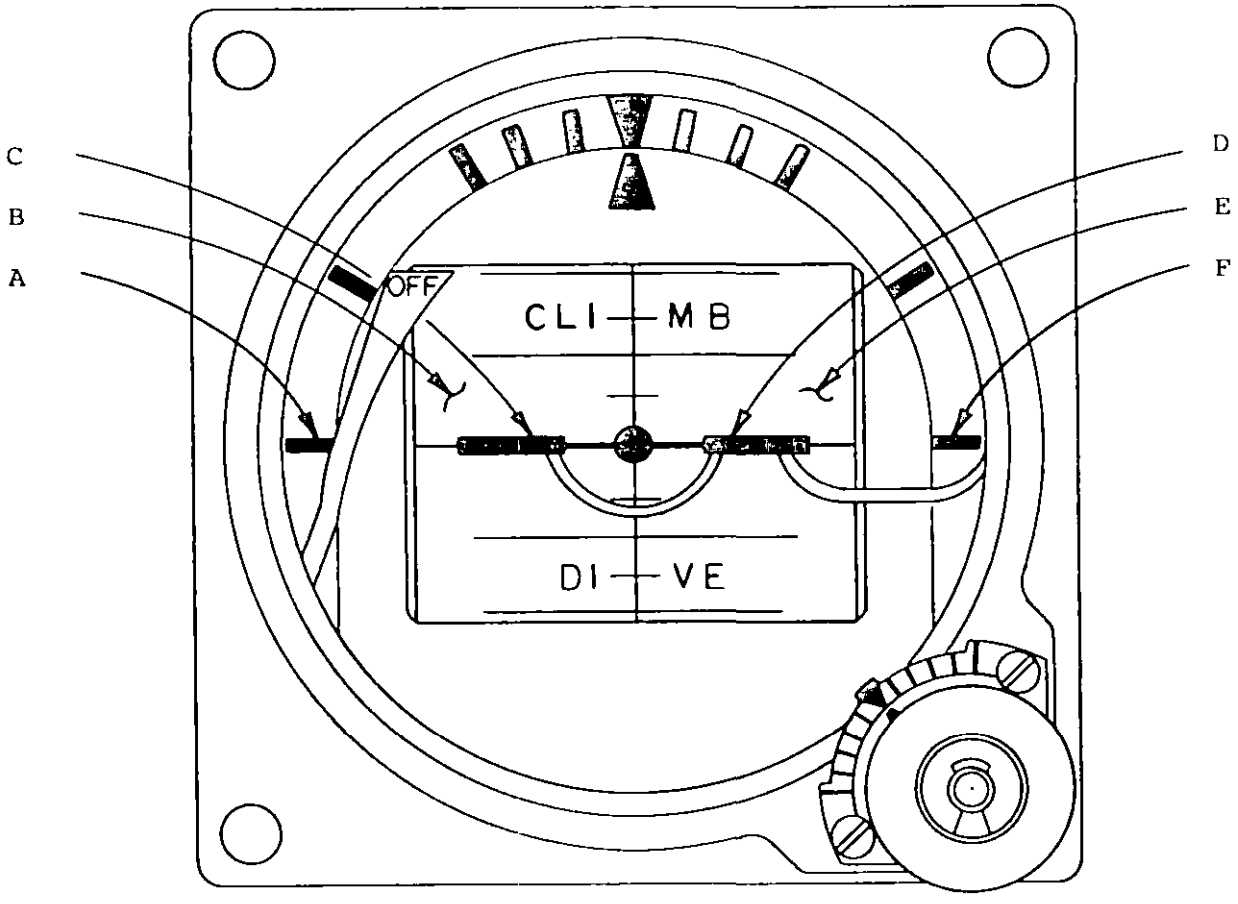


FIGURE 2 INDICATOR DISPLAY

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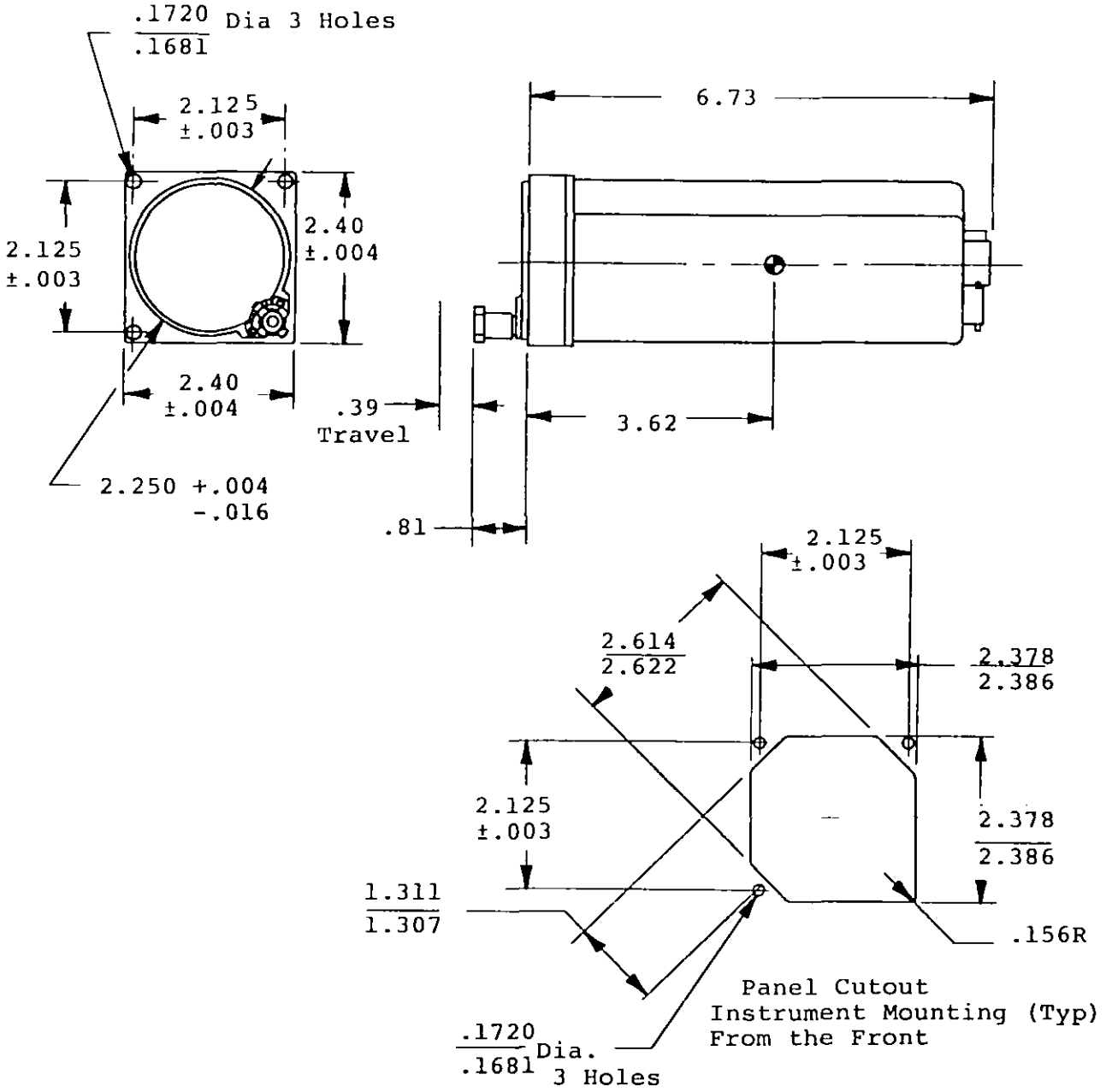
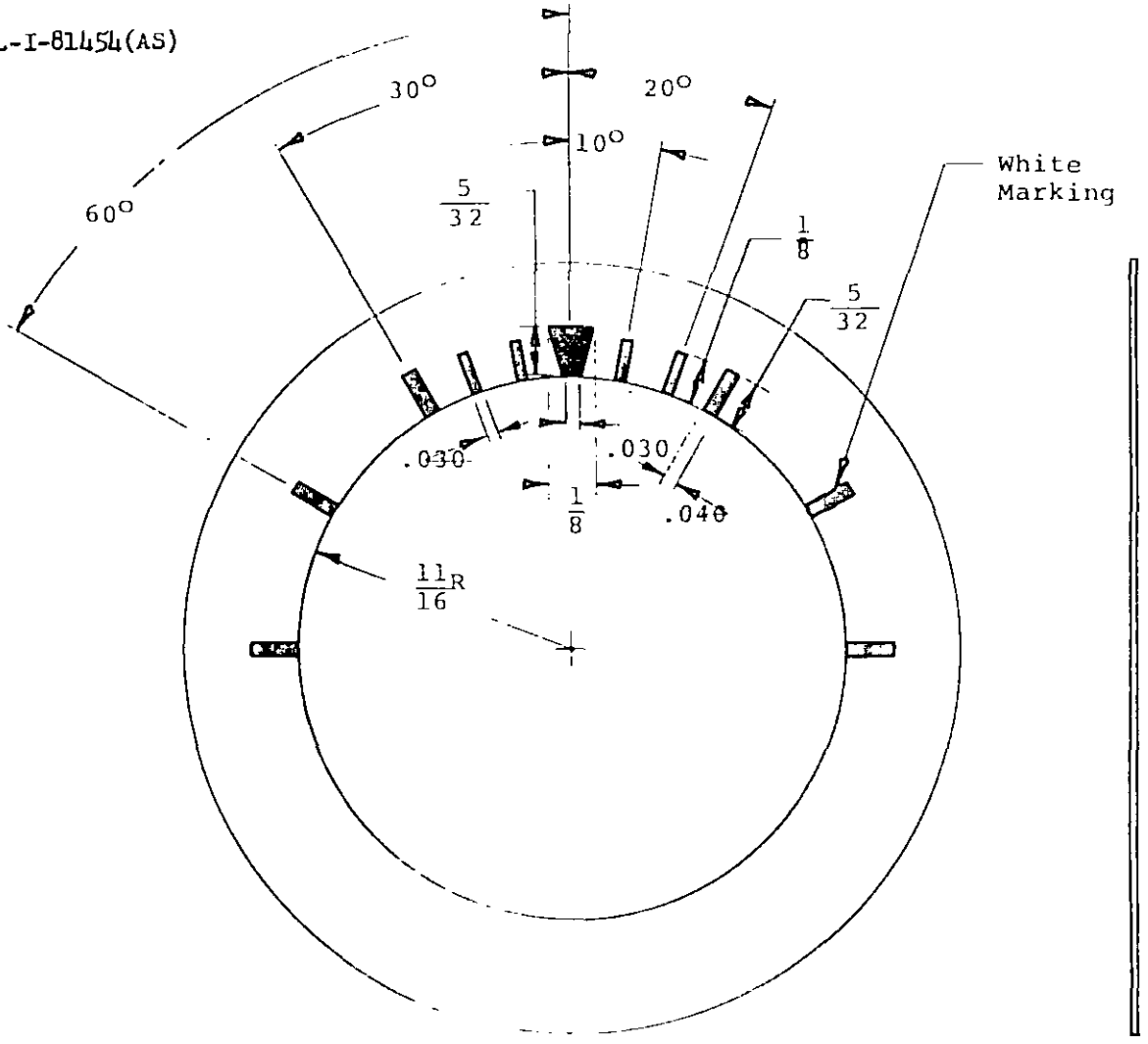


FIGURE 3 OUTLINE DIMENSIONS

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DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED
TOLERANCES ARE:
FRACTIONS ± 1/64
DECIMALS ± .005
ANGLES ±1°

FIGURE 4 BANK ANGLE DIAL (TOP)

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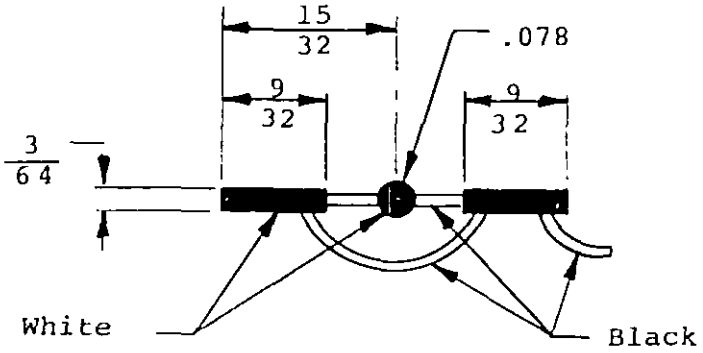


FIGURE 6 MINIATURE AIRPLANE

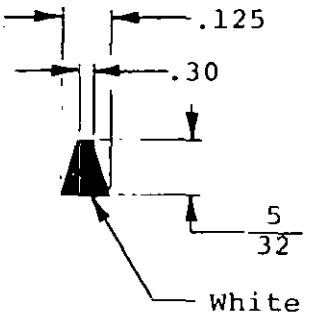
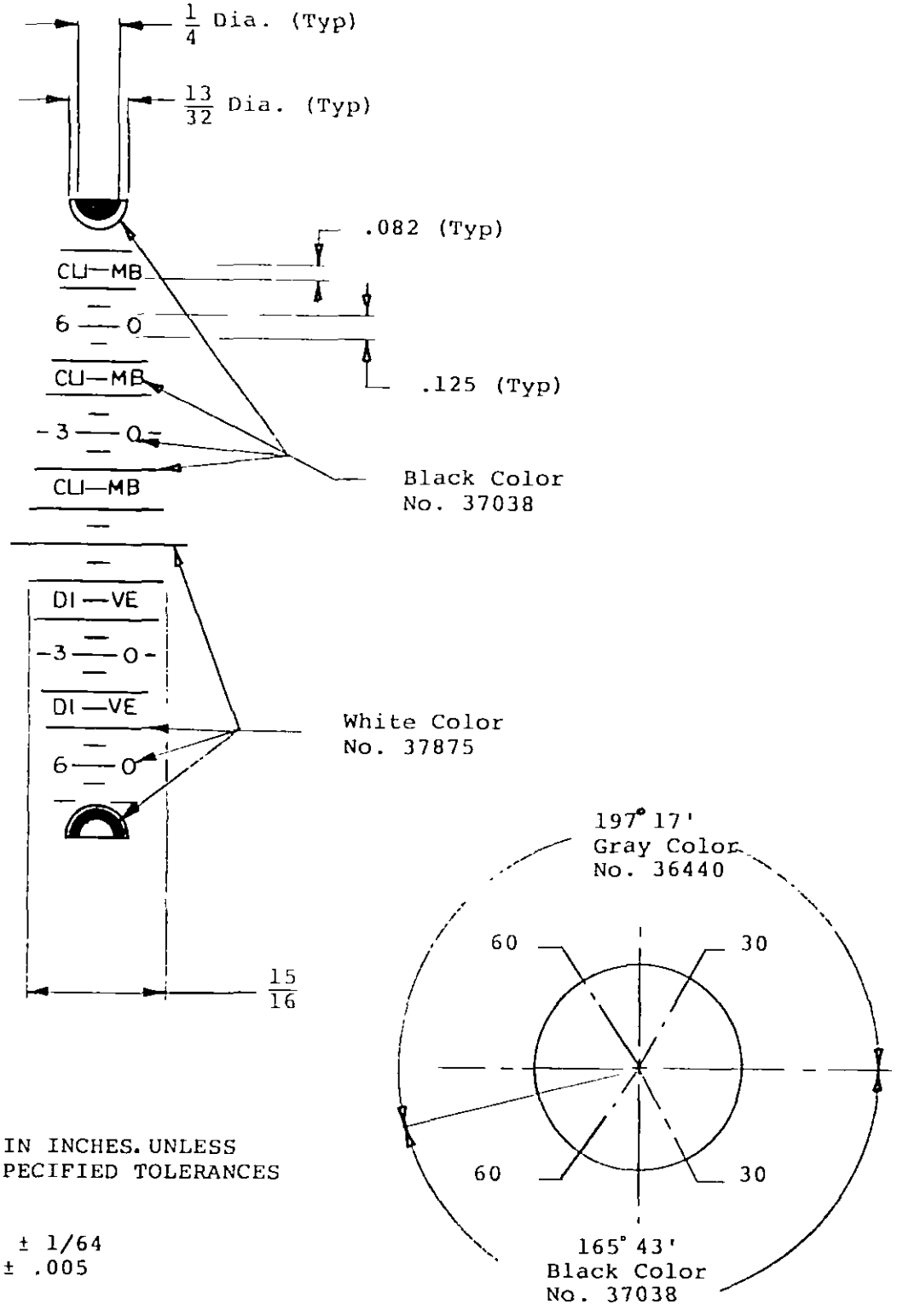


FIGURE 5 BANK INDEX

DIMENSIONS IN INCHES. UNLESS
OTHERWISE SPECIFIED TOLERANCES
ARE:

FRACTIONS ± 1/64
DECIMALS ± .005

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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED TOLERANCES ARE:

FRACTIONS ± 1/64
DECIMALS ± .005

FIGURE 7 DIMENSIONS AND MARKINGS

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-I-81454(AS) Indicator, Gyroscope, Vertical Reference ID-1481(*)/A

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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Naval Air Systems Command
Washington, D.C. 20360

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